

MASTER COST REPORT

CostVision · Comprehensive Analysis Package

Generated: 07 Aug 2026

Parts Loaded: Should-Cost

Report Contents

- PART A** Should-Cost Analysis (\$A1–\$A5: breakdown, operations, DFM/DFA, optimisations)
- PART B** AI CAD-to-Cost Analysis (\$B1–\$B5: geometry, process recs, materials, risks) [not loaded]
- PART C** PCB Image Analysis (\$C1–\$C6: board spec, assembly, BOM, country comparison) [not loaded]

Should-Cost Summary

Part: Bosch IPB2.0 ECU PCBA
Total: £112.34 · Material: 70% · Process: 13% · Labour: 1% · Tooling NRE: —

PART A — SHOULD-COST ANALYSIS

Bosch IPB2.0 ECU PCBA · PCBA · CN · 07 Aug 2026

Manufacturing Basis

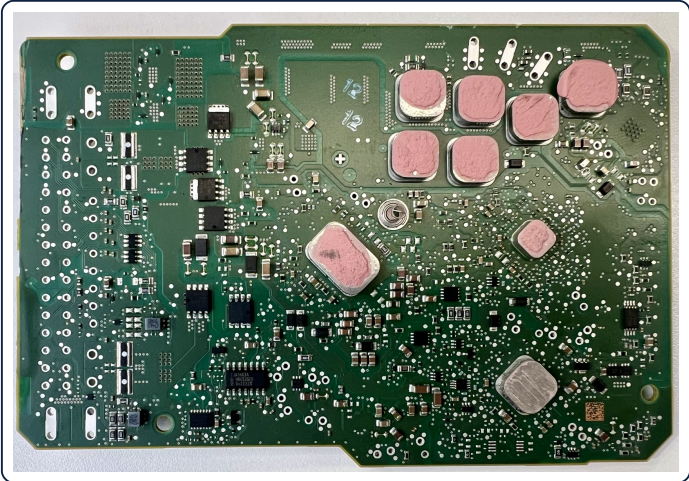
Region: CN · Currency: GBP · Annual volume: 200,000 · Overhead 9% · Margin 8%

Source Photographs

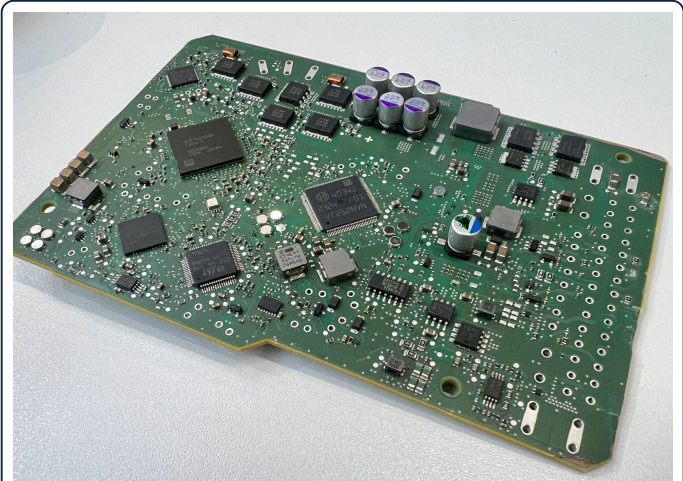
3 images the costing was built from

Every component identification in the bill of materials traces to a package marking legible in one of these images. Lines flagged as estimated could not be read from any of them.

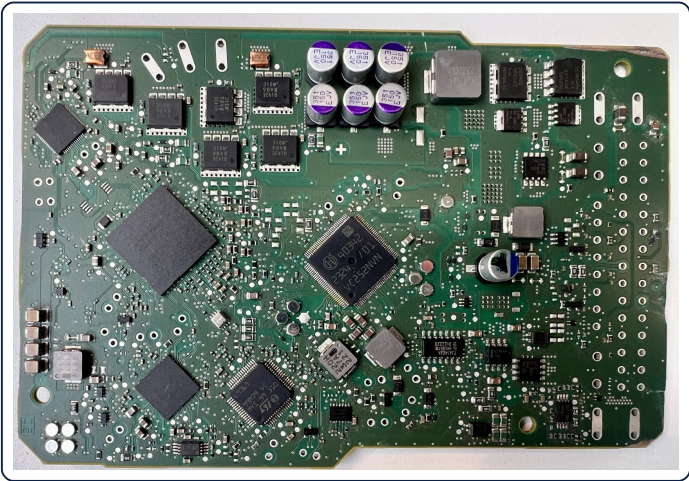
Top side



Bottom side



Close-up 1



§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (GBP)	% of Total	Cost Mix Bar
1. Raw Material	£78.55	69.9%	
2. Process (Machine)	£14.21	12.6%	
3. Direct Labour	£1.25	1.1%	

Cost Bucket	Amount (GBP)	% of Total	Cost Mix Bar
4. Tooling (amortised)	£0.23	0.2%	
5. Packaging	£0.73	0.6%	
6. Logistics	£0.57	0.5%	
Factory Cost	£95.53	85.0%	
7. Overhead (SG&A)	£8.48	7.5%	
Subtotal	£104.01	92.6%	
8. Supplier Margin	£8.32	7.4%	
TOTAL SHOULD-COST	£112.34	100.0%	

\$2 — Commercial Parameters

Overhead Rate	9.0%	Supplier Margin	8.0%
Packaging / part	£0.73	Logistics / part	£0.57
Total Tooling Cost	£46000.00	Amortisation Volume	200,000 parts

Functional Safety (ISO 26262)

grade assumed - not yet confirmed

Parameter	Value	Basis
Quality grade costed	auto grade1	Sets the multiplier on test and inspection cycle times
Test / inspection multiplier	x1.80	Applied to every test and inspection operation in section 4
Test & inspection cost (2 ops)	£6.43	BGA X-Ray Inspection (inline AXI), ICT / Functional Test
Same routing at consumer grade	£3.57	The safety grade adds £2.86/part to verification
ASIL determination	NOT SUPPLIED	An ASIL comes from hazard analysis (HARA), not from a board photograph

No ASIL has been supplied for this costing, so the figures above record the quality grade that was ASSUMED rather than a safety rating this tool determined. Confirm the grade against the item safety requirement before quoting: a step up or down changes the verification cost shown above roughly in proportion to the multiplier, and can change component selection and the X-ray strategy as well.

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-virtual	ID	
Grade / Specification	Virtual / Pass-through		Placeholder for directCost modules (painting, BIW, PCB, PCBA). Price is irrelevant — directCost overrides. Regional adj. ×0.880 (other)
Region	China		
Net Finished Weight	0.0000 kg	kg	Weight in finished part
Direct Material Cost	£78.55	GBP	Bypasses weight-based model

Data Confidence	Medium
Effective Date	2026-06-14

3B Bill of Materials (20 lines · 731 pieces)

#	RefDes	Description	Pkg	Value	Qty	Unit GBP	Ext GBP
1	U-MCU	Renesas R7F702300B RH850/U2A 292-BGA MARKING LEGIBLE - "R7F702300B FABA-C BB05253 2308 JAPAN" fully legible. DigiKey \$49.47@1.	BGA-292		1	£14.02	£14.02
2	U-SBC	ST L9369 dual H-bridge EPB pre-driver MARKING LEGIBLE - "L9369 VC DTE" + ST logo, LQFP-64. Confirms this is an EPB/brake ECU.	LQFP-64		1	£2.41	£2.41
3	U-CAN	NXP TJA1463AT CAN-FD SIC transceiver MARKING PARTIAL - CORRECTED: board shows SO-14 gull-wing = TJA1463AT. Previously priced from TJA1463ATK, which is HVSON-14 - wrong package. Qty 2 is an assumption; only one is clearly visible.	SO-14		2	£0.58	£1.17
4	U-NCV	onsemi NCV8461 protected high-side switch MARKING LEGIBLE - "NCV8461 PR17" legible. Qty CUT 2 -> 1: only one instance is visible.	SOIC-8		1	£0.29	£0.29
5	U-UNK1	UNIDENTIFIED 8-pin, marking "T698 / 1000T / " MARKING PARTIAL - Marking visible but not resolvable to a manufacturer. Appears at least twice. Class median.	SOIC	2809h	2	£0.33	£0.67
6	U-UNK2	UNIDENTIFIED small package, marking "CH40 3095 / 9096" MARKING PARTIAL - Two similar parts, marking legible, part not identifiable.	SOIC		2	£0.17	£0.33
7	U-40342	Bosch 40342/01 motor-control ASIC MARKING LEGIBLE - "40342 /01 2324 VC252NVN" + Bosch logo.	QFP-144		1	£3.50	£3.50
8	U-40341	Bosch 40341/01 ASIC MARKING LEGIBLE - "40341 /01 2240 A2I92F4A" + Bosch logo.	QFP-100		1	£2.50	£2.50

#	RefDes	Description	Pkg	Value	Qty	Unit GBP	Ext GBP
9	U-23027	Bosch 2302701 ASIC MARKING PARTIAL - "2302701" + Bosch logo legible; the line above it ("SSEAC"?) is not resolved.	QFP-64		1	£1.80	£1.80
10	Q1-6	Bosch Q142E / 0142E power stage MARKING PARTIAL - AMBIGUOUS: leading character reads Q in some views and 0 in others. Suffix "BABA .W31C". Qty 6 counted from one close-up; more may sit outside frame.	PowerSO-8		6	£1.20	£7.20
11	U-71H	Bosch 71H740 driver / power-path MARKING LEGIBLE - "71H740 Prm 2321 C3 5371" legible. Qty 3 is a count across overlapping views - may double-count.	SOIC-8		3	£0.65	£1.95
12	U-76E2	Bosch 76E240 device MARKING LEGIBLE - "76E240 PNW 2328 B6 0125" legible. Qty 3 same caveat as 71H740.	SOIC-8		3	£0.70	£2.10
13	U-76E8	Bosch 76E840 device MARKING LEGIBLE - "76E840 PEm 2319 A2 1431" legible.	SOIC-8		1	£0.70	£0.70
14	U-7S1R	Bosch 7S1R540H power device MARKING LEGIBLE - "7S1R540H Prt" x2, lot codes 2313D6 and 026200.	SOIC-8		2	£0.80	£1.60
15	C-BULK	Polymer alu capacitor, marking "351 150 EJV" MARKING PARTIAL - Marking legible; /35V is an INTERPRETATION of it, not a datasheet reading.	0805	150uF	6	£0.29	£1.74
16	L-PWR	Shielded power inductors / 10uH / 22uH / 220H MARKING PARTIAL - Four DISTINCT values readable (0.47uH 23220J, 10uH, 22uH 2316NJ, 220H 374A) - previously lumped as one line at one price.	Power module	0.47uH	8	£0.38	£3.03
17	R/C	MLCC + thick-film resistors (1201, 1001, 7501, 000, 30.9, 120E, 4021, 100E, 10DE) INFERRED - Individual values readable in close-ups; the COUNT of 600 is a density estimate over both sides, not a count.	0402		600	£0.01	£3.66
18	D/Q	Small-signal semis + BRL-23/BRL-97 diodes INFERRED - "BRL 23 M", "BRL 97 M", "WJY", "52s", "Y7A A00A 110", "CXR 42AA" all visible. Count estimated.	SOT/SOD		80	£0.04	£3.11
19	J1-2	Automotive press-fit headers (main + motor) INFERRED - Press-fit hole fields visible along two board edges; connectors themselves not in frame.	THT		2	£1.78	£3.56
20	TIM	Thermal interface pads (both sides) MARKING LEGIBLE - Pink pads clearly visible bottom side, grey/graphite pads top side.	THT		8	£0.08	£0.67
BOM TOTAL					731		£56.00

Lines total £56.00 against a £78.55 material bucket - the £22.55 balance is carried outside the itemisation (bare board, yield/rework allowance, coating).

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
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Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OOE %	Machine Cost
SMT Placement & Reflow	High-Speed SMT Line (Fuji/Juki/ASM)	£78.38	5.28	1	82.0%	£8.41
TH Insertion & Hand Soldering	Assembly Workbench	£0.36	0.40	1	100.0 %	£0.00
BGA X-Ray Inspection (inline AXI)	X-Ray BGA Inspection Cell (2D/3D AXI)	£47.63	1.35	1	90.0%	£1.19
ICT / Functional Test	ICT Bed-of-Nails Test System (Automotive)	£58.39	4.50	1	95.0%	£4.61
TOTAL						£14.21

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
SMT Placement & Reflow	SMT / Electronics Operator	£6.50	1	5.28	100.0%	£0.57	£8.98
TH Insertion & Hand Soldering	SMT / Electronics Operator	£6.50	1	0.40	100.0%	£0.04	£0.05
BGA X-Ray Inspection (inline AXI)	SMT / Electronics Operator	£6.50	1	1.35	100.0%	£0.15	£1.34
ICT / Functional Test	SMT / Electronics Operator	£6.50	1	4.50	100.0%	£0.49	£5.10
TOTAL						£1.25	£15.46

\$5 — Machine Rate Buildup

Assembly Workbench [bench-assembly] · Confidence: Low · Assembly bench — low machine rate; cost primarily driven by labour | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @85% util	Notes
Depreciation	£275.00	£0.08	4,000 hr/yr available
Maintenance	£110.00	£0.03	
Energy	£91.30	£0.03	
Floor Space	£440.00	£0.13	
Indirect Support	£275.00	£0.08	
Finance Cost	£27.50	£0.01	
TOTAL — Assembly Workbench	£1218.80	£0.36	Effective hours: 3400/yr

High-Speed SMT Line (Fuji/Juki/ASM) [smt-high-speed-line] · Confidence: Low · High-speed SMT line 80000+ CPH. Automotive EMS benchmark. Target £150/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @82% util	Notes
Depreciation	£99000.00	£30.18	4,000 hr/yr available
Maintenance	£44000.00	£13.41	
Energy	£16739.13	£5.10	
Floor Space	£16500.00	£5.03	
Indirect Support	£55000.00	£16.77	
Finance Cost	£25850.00	£7.88	
TOTAL — High-Speed SMT Line (Fuji/Juki/ASM)	£257089.13	£78.38	Effective hours: 3280/yr

X-Ray BGA Inspection Cell (2D/3D AXI) [xray-bga-inspection] · Confidence: Low · 2D/3D automated X-ray inspection for BGA solder joints. EMS automotive benchmark. Target £90/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @82% util	Notes
Depreciation	£66000.00	£20.12	4,000 hr/yr available
Maintenance	£24750.00	£7.55	
Energy	£7608.70	£2.32	
Floor Space	£11000.00	£3.35	
Indirect Support	£35750.00	£10.90	
Finance Cost	£11110.00	£3.39	
TOTAL — X-Ray BGA Inspection	£156218.70	£47.63	Effective hours: 3280/yr

ICT Bed-of-Nails Test System (Automotive) [ict-automotive] · Confidence: Low · In-circuit test fixture for IATF 16949 automotive boards. Target £110/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @82% util	Notes
Depreciation	£71500.00	£21.80	4,000 hr/yr available
Maintenance	£30250.00	£9.22	
Energy	£9130.43	£2.78	
Floor Space	£13750.00	£4.19	
Indirect Support	£46750.00	£14.25	
Finance Cost	£20130.00	£6.14	
TOTAL — ICT Bed-of-Nails Test System (Automotive)	£191510.43	£58.39	Effective hours: 3280/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

\$6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
rawMaterial.directCost	78.5453	£	Pre-computed by commodity module	mat-virtual	Medium
SMT Placement & Reflow.machineRatePerHr	78.3808	£/hr	High-speed SMT line 80000+ CPH. Automotive EMS benchmark. Target £150/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	smt-high-speed-line	Low
SMT Placement & Reflow.labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low
TH Insertion & Hand Soldering.machineRatePerHr	0.3585	£/hr	Assembly bench — low machine rate; cost primarily driven by labour Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	bench-assembly	Low
TH Insertion & Hand Soldering.labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low
BGA X-Ray Inspection (inline AXI).machineRatePerHr	47.6277	£/hr	2D/3D automated X-ray inspection for BGA solder joints. EMS automotive benchmark. Target £90/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	xray-bga-inspection	Low
BGA X-Ray Inspection (inline AXI).labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low

Field	Value	Unit	Source / Reference	Rate ID	Conf.
ICT / Functional Test.machineRatePerHr	58.3873	£/hr	In-circuit test fixture for IATF 16949 automotive boards. Target £110/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	ict-automotive	Low
ICT / Functional Test.labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
£112.34 ± 23.2%	£87.82	£110.04	£140.02	wide	18.6%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 8 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread.

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.

Import duty and international freight (regional table is Ex-Works).

Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by £ impact

Cost Driver	Base	-10% cost	+10% cost	Range £
Direct material cost	£78.55	-8.2%	+8.2%	£18.49
SMT Placement & Reflow: Cycle Time	0.09hr	-0.9%	+0.9%	£2.11
SMT Placement & Reflow: Machine Rate	£78.38/hr	-0.9%	+0.9%	£1.98
Overhead %	9%	-0.8%	+0.8%	£1.83
Supplier Margin %	8%	-0.7%	+0.7%	£1.66
ICT / Functional Test: Cycle Time	0.08hr	-0.5%	+0.5%	£1.20
ICT / Functional Test: Machine Rate	£58.39/hr	-0.5%	+0.5%	£1.09
BGA X-Ray Inspection (inline AXI): Cycle Time	0.02hr	-0.1%	+0.1%	£0.31

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	£89.26	£25.84	£4.50	£0.42	£11.31	£131.31	£0.39	£141.07	+26%
Germany (EUR)	£91.93	£27.13	£7.27	£0.44	£13.14	£139.90	£0.45	£150.32	+34%
France (EUR)	£91.04	£23.77	£5.22	£0.38	£11.91	£132.33	£0.45	£142.27	+27%
Spain (EUR)	£89.26	£22.74	£3.29	£0.37	£10.35	£126.00	£0.45	£135.51	+21%
Poland (PLN)	£86.58	£18.60	£2.04	£0.30	£8.61	£116.14	£0.47	£124.91	+11%
Turkey (TRY)	£80.33	£15.50	£1.14	£0.25	£7.15	£104.36	£0.51	£112.33	-0%
China (CNY)	£78.55	£14.21	£1.25	£0.23	£8.48	£102.71	£0.57	£112.34	Base
India (INR)	£80.33	£13.43	£0.79	£0.22	£6.43	£101.21	£0.59	£108.90	-3%
Mexico (MXN)	£84.79	£15.50	£1.32	£0.25	£7.49	£109.35	£0.53	£117.60	+5%
United States (USD)	£89.26	£21.96	£5.90	£0.36	£10.52	£127.99	£0.49	£137.59	+22%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

0.00 kgCO2e · 0.00 /kg · cradle-to-gate

Material	0.00 kgCO2e	Material factor	3.00 kg/kg (Generic (assumed))
Process	0.00 kgCO2e	Grid intensity	0.550 kg/kWh · 0.00 kWh
Logistics	0.00 kgCO2e	Total	0.00 kgCO2e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[WARNING] Low OEE on "SMT Placement & Reflow"

Finding	OEE of 82% is below the 90% benchmark. Every 5% OEE improvement on this operation saves ~£0.710/part.
Impact	Medium · up to 1% potential saving
Action 1	Implement SMED (Single-Minute Exchange of Die) to reduce changeover losses
Action 2	Schedule preventive maintenance to reduce unplanned downtime

§12 — Cost-Reduction Opportunities (ranked by saving)

4 opportunities · combined £7.64/part (~7%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Design & Geometry	Design Review for DFM Compliance	Formal DFM review with manufacturing engineering to identify part features that increase cost without functional benefit.	£5.62	5.0%	Quick Win	Low
2	Quality & Inspection	Right-Size Inspection and Test	Inspection/test operations carry 5.7% of part cost as separate steps. Capable processes earn reduced inspection — that is what capability data is for.	£3.86	3.4%	Medium Term	Low
3	Commercial & Sourcing	Raw-Material Price Indexation Clause	With material at 69.9% of cost, the quote embeds the supplier's hedge against metal/resin volatility. An index-linked clause removes that risk premium from the piece price.	£3.37	3.0%	Quick Win	Low
4	Commercial & Sourcing	Annual Volume Re-Commitment for Better Pricing	Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	£3.37	3.0%	Quick Win	Low

§13 — Opportunity by Category

3 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Design & Geometry	1	Design Review for DFM Compliance	£5.62	£5.62
Quality & Inspection	1	Right-Size Inspection and Test	£3.86	£3.86
Commercial & Sourcing	2	Raw-Material Price Indexation Clause	£3.37	£6.74

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Raw-Material Price Indexation Clause
- Design Review for DFM Compliance
- Annual Volume Re-Commitment for Better Pricing